



JAVA PROJECT

PRESENTATION

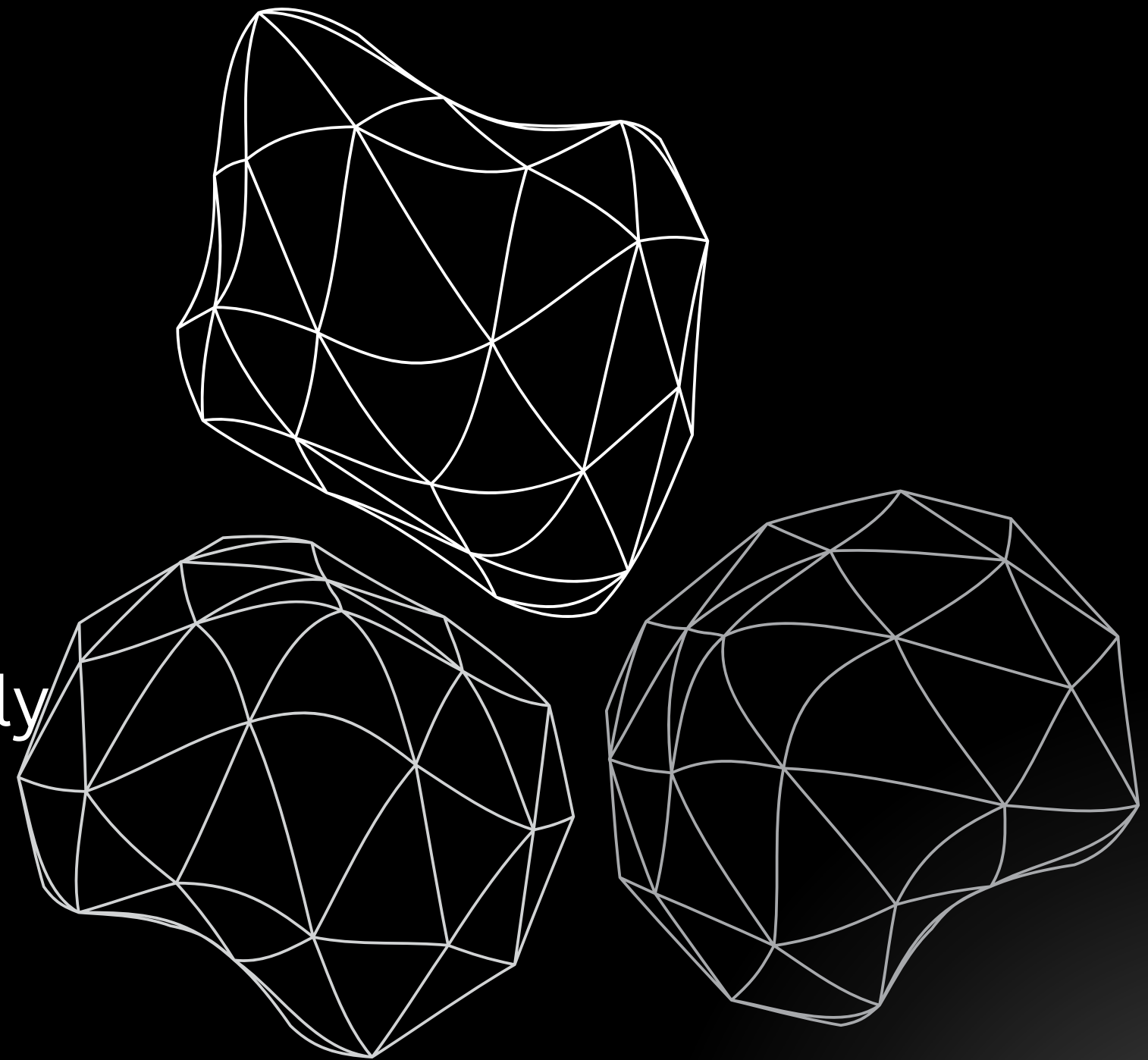
PRESENTED BY AYUSH KUMAR SINGH

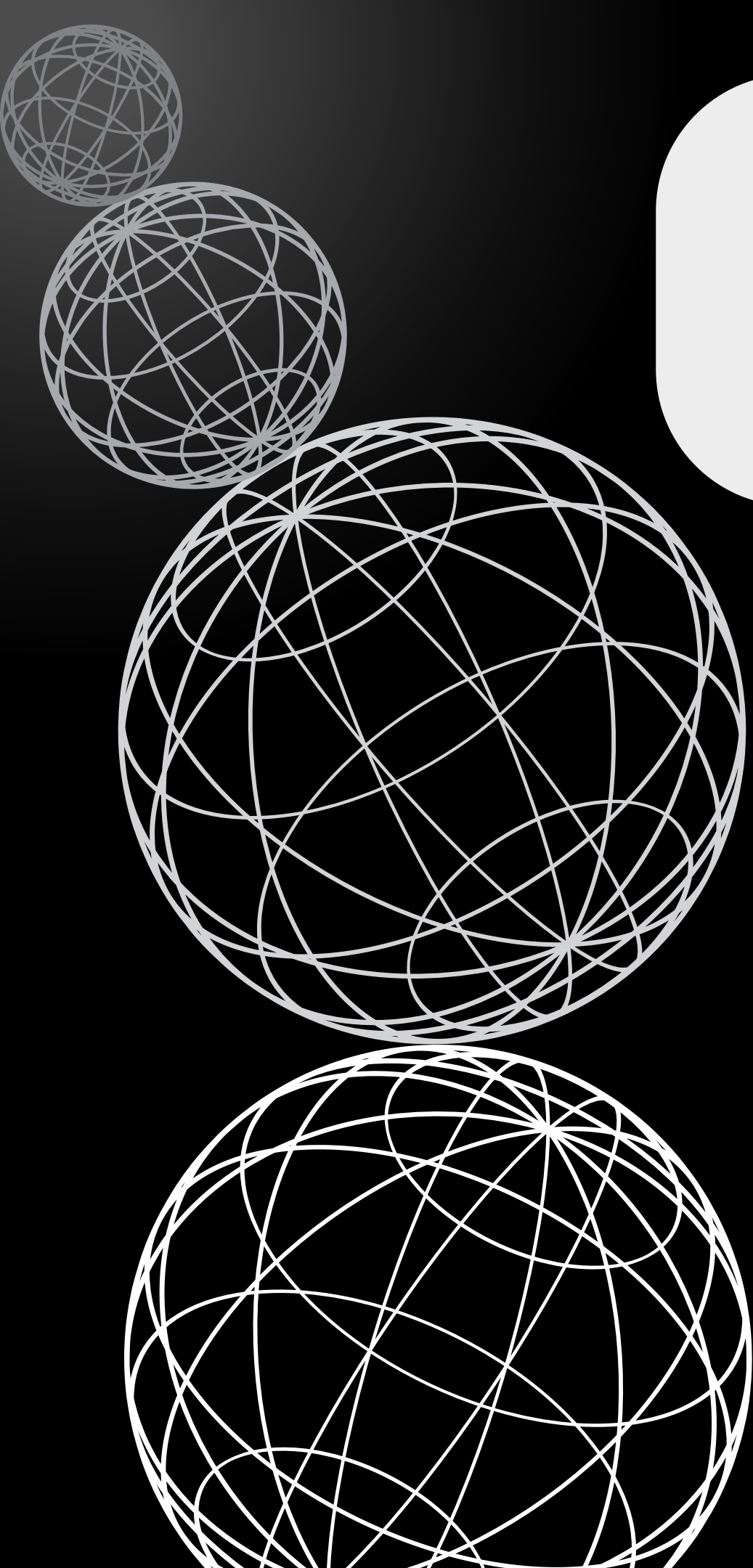
INTRODUCTION

The Flight Reservation System is a Java-based application developed using Object-Oriented Programming concepts.

PURPOSE

- Manage passenger bookings efficiently
- Automate seat reservation process
- Prevent duplicate bookings
- Display seat availability in real-time



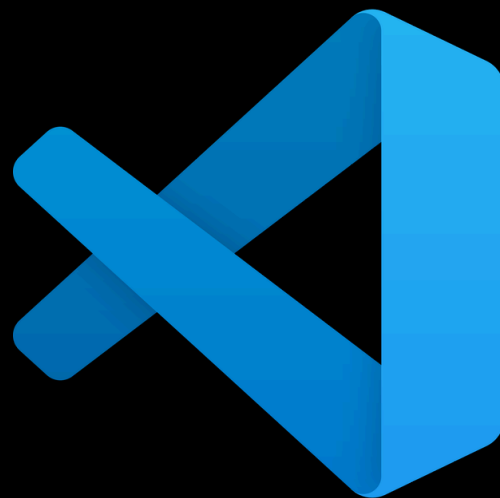


TECHNOLOGIES USED

- Core Java
- OOP Concepts
- HTML, CSS, JavaScript (Frontend)
- Java HTTP Server (Backend)



OOP CONCEPTS USED

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- Classes & Objects
 - Encapsulation
 - Constructors
 - Arrays / ArrayList
 - HAS-A Relationship

FEATURES

PASSENGER MANAGEMENT

- Add Passenger
- Search Passenger
- View Passenger Detail

SEAT RESERVATION

- Book Seat
- Cancel Reservation
- Check Available Seats

VALIDATION

- Prevent Duplicate Passenger IDs
- Check Seat Availability
- Input Validation

USER INTERFACE

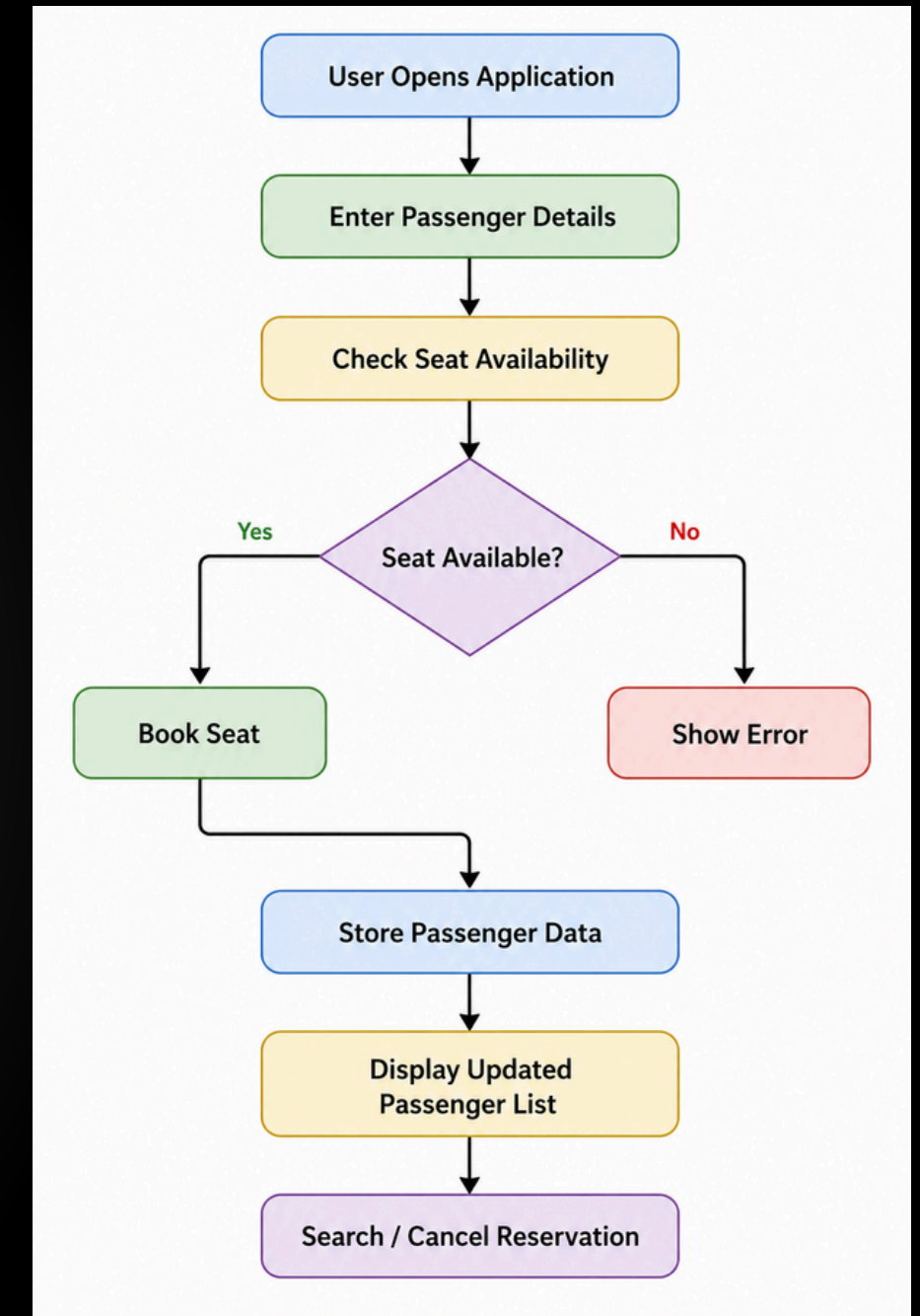
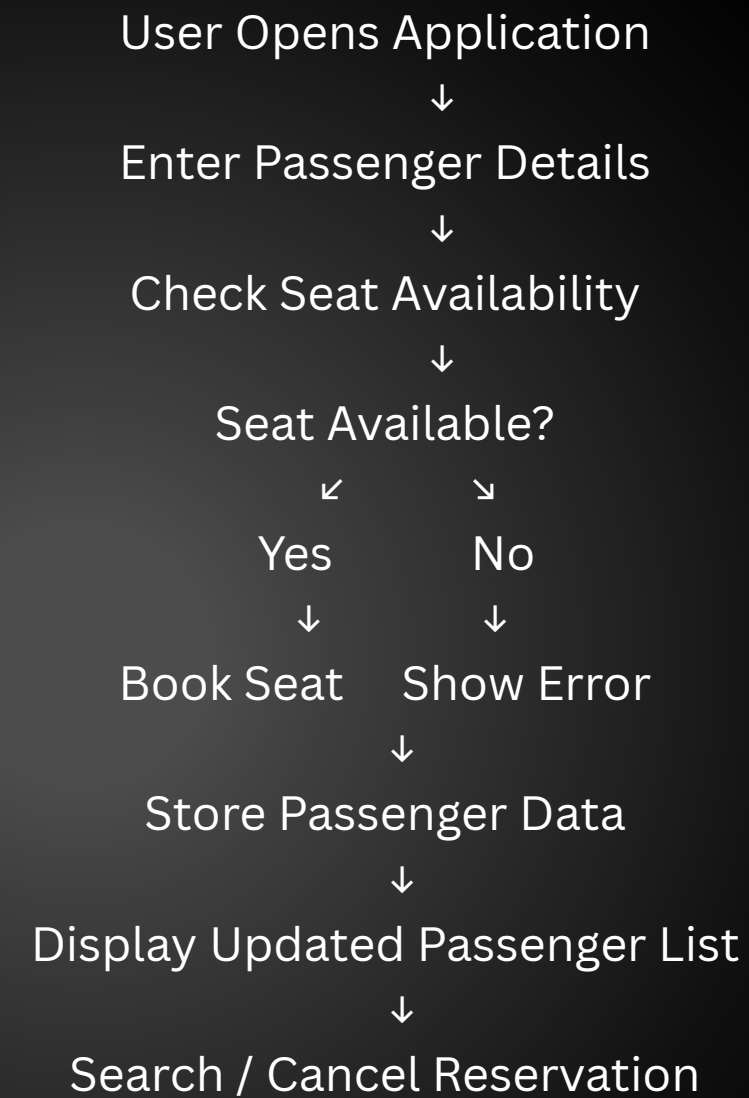
- Responsive Design
- Simple Dashboard
- Real-time Updates

WORKING OF THE PROJECT

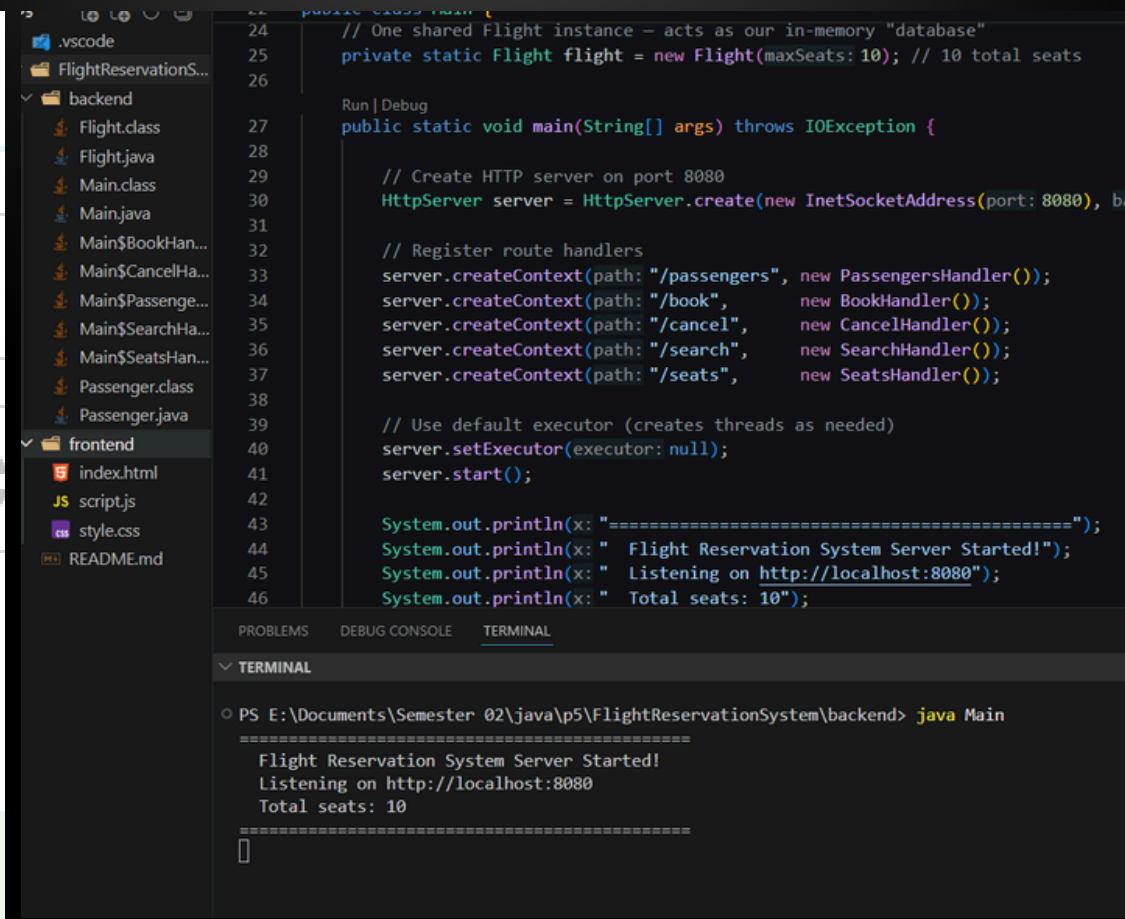
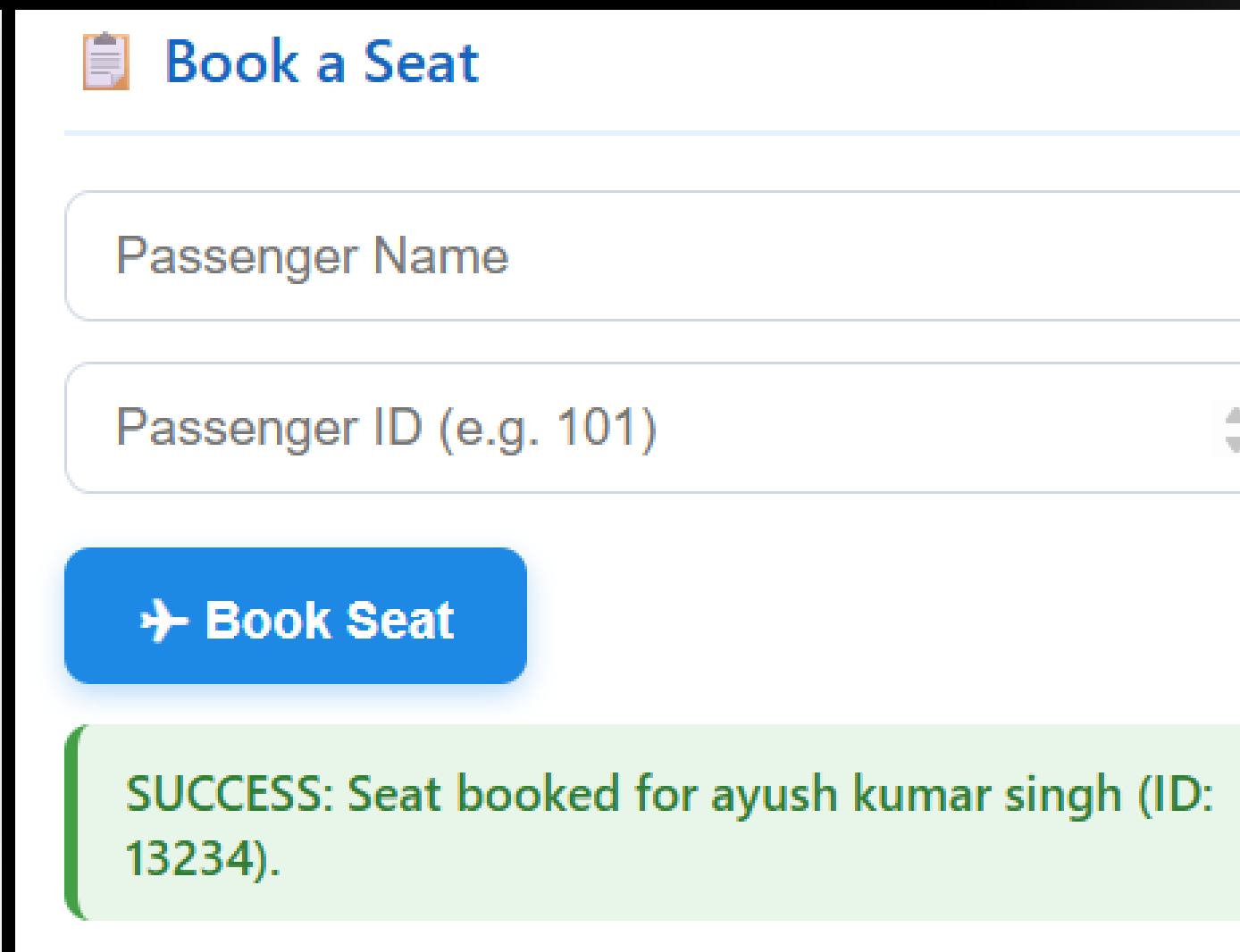
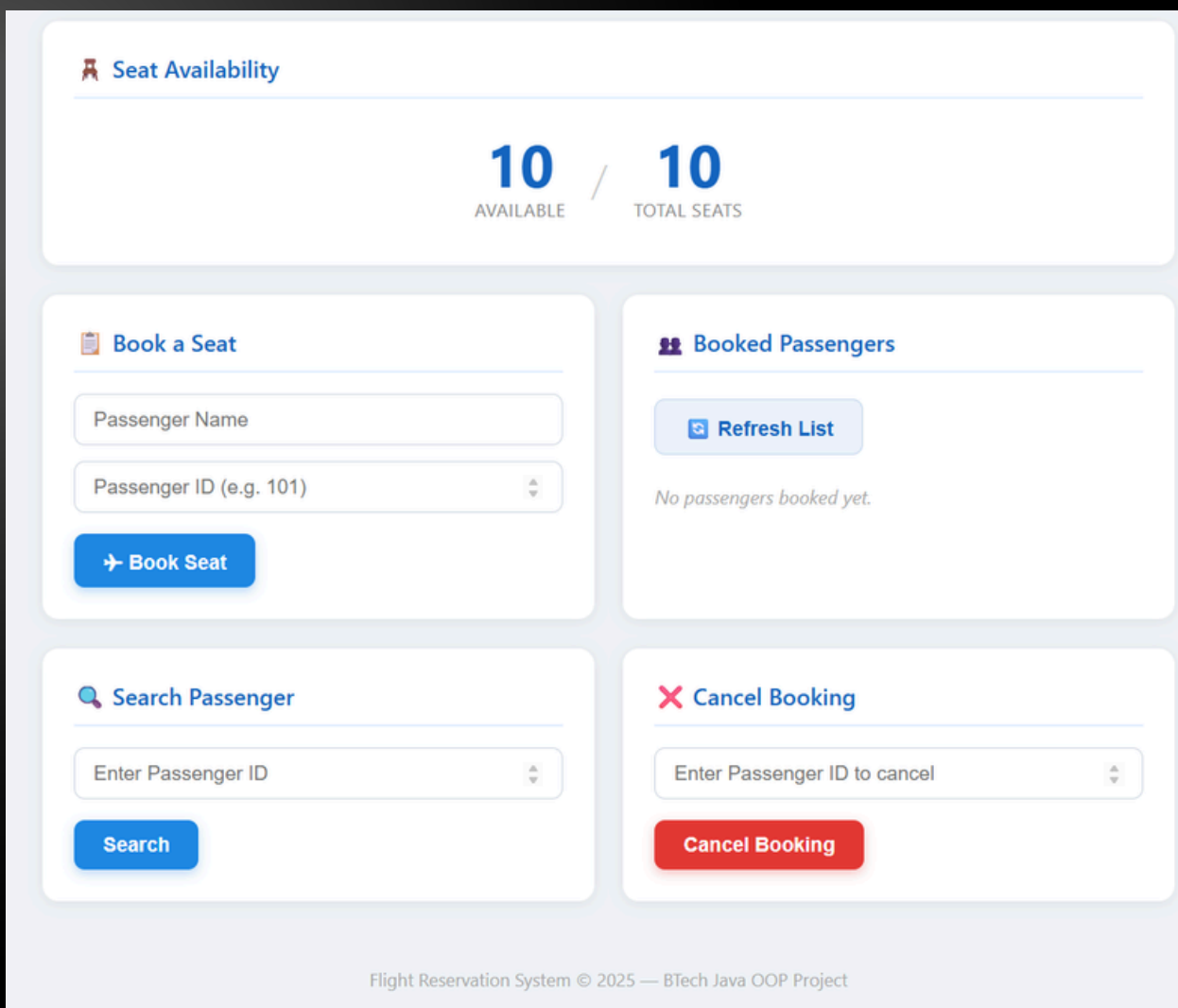
SEARCH / CANCEL RESERVATION

Backend Process

- 1.Create Flight Object
- 2.Create Passenger Object
- 3.Check Duplicate ID
- 4.Book Seat
- 5.Store in ArrayList
- 6.Display Current Status



SCREENSHOTS OF THE PROJECT





FUTURE SCOPE

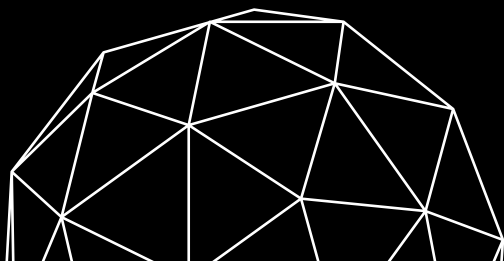
TECHNICAL IMPROVEMENTS

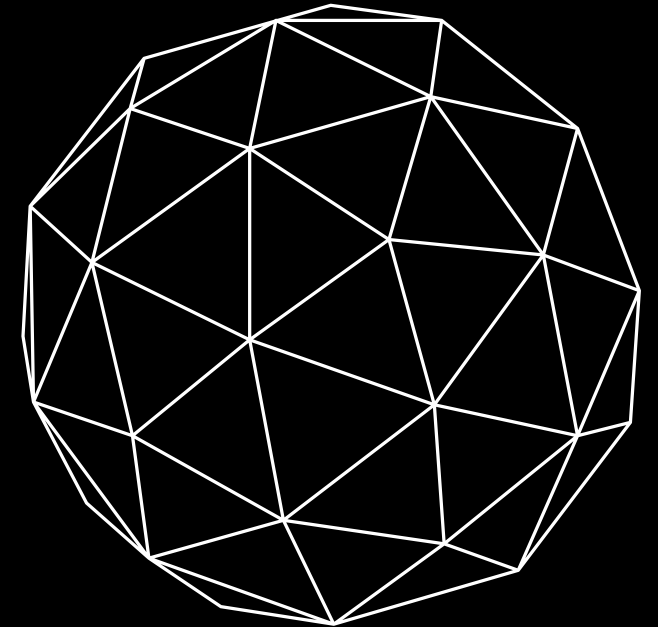
- Database Integration (MySQL)
- Login & Authentication
- Ticket Generation
- Online Payment Gateway
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ADVANCED FEATURES

- Seat Selection
- Multiple Flights
- Flight Schedule Management
- Email/SMS Confirmation
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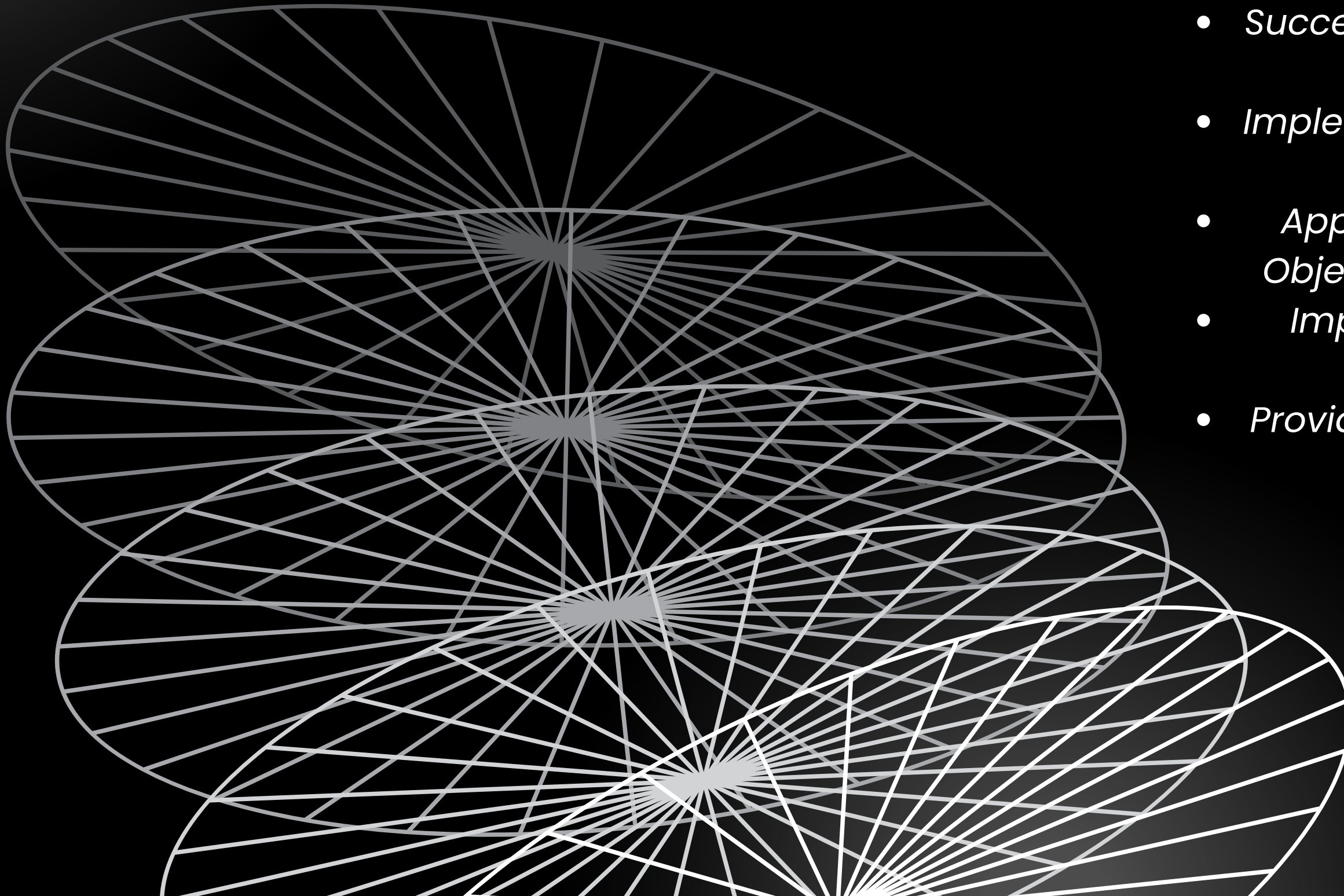
UI ENHANCEMENTS

- Java Swing GUI
 - Mobile-Friendly Interface
 - Dark Mode Support
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CONCLUSION

- *Successfully developed a Flight Reservation System using Java OOP.*
- *Implemented passenger management and seat booking functionalities.*
- *Applied OOP concepts such as Classes, Objects, Encapsulation, and Constructors.*
- *Improved understanding of real-world software development.*
- *Provides a strong foundation for advanced airline reservation systems.*



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THANK YOU